

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

11th September 2021

**Subject CM2A – Financial Engineering and Loss
Reserving (Paper A)**

Time allowed: 3 Hours 30 Minutes (09.30 – 13.00 Hours)

Total Marks: 100

Q. 1) i) Which of the following is correct?

- I.** X_1 follows normal distribution, X_2 follows normal distribution, then $X_1 * X_2$ follows lognormal distribution
- II.** X_1 follows normal distribution, X_2 follows normal distribution, then $X_1 + X_2$ follows lognormal distribution
- III.** X_1 follows lognormal distribution, X_2 follows lognormal distribution, then $X_1 + X_2$ follows lognormal distribution
- IV.** X_1 follows lognormal distribution, X_2 follows lognormal distribution, then $X_1 * X_2$ follows lognormal distribution (2)

ii) An analyst has realised that Beta estimated for a stock is unsuitable for any kind of prediction. Which of the following could be the reasons for the same?

- a. Assumptions of CAPM does not hold
- b. the company got into a risky venture
- c. Data was not sufficient
- d. the stock got merged into another company
- I.** all of the above
- II.** none of the above
- III.** a, b and c
- IV.** a, c and d (2)

iii) A company declares a cash dividend unexpectedly, what would be immediate effect on the value of call option and put option price on the company's share.

- a. The value of Call option will increase
- b. The value of Put option will increase
- c. The value of Call option will remain unchanged
- d. The value of Call option will decrease
- I.** both a and b
- II.** only c
- III.** both b and d
- IV.** only d (2)

iv) Which of the following (is) are incorrect?

- a. the variance of the return on the portfolio gets closer to the average covariance of return between the pairs of assets in that portfolio.
- b. As a portfolio is diversified, the return on the portfolio is less exposed to the specific risk of any one component.
- c. as portfolios are diversified the correlation components become less important
- d. as portfolios are diversified the correlation components become more important
- I.** only c
- II.** b and c
- III.** only d
- IV.** a and b (2)

v) Which of the following class of people can use Mean –variance portfolio theory?

- a. A mutual fund which can only buy stocks
- b. A risk –averse investor
- c. An hedge fund which can buy stocks or short sell stocks
- d. A risk neutral investor

- I. only a
- II. only d
- III. b and c
- IV. a and b

(2)

[10]

Q. 2) i) Give one word on which form of efficient markets hypothesis is represented by each of following statements:

- I. It should not be possible to consistently profit by selling winners and hanging on to losers.
- II. It should not be possible to consistently profit by trading on any public information, such as that found on the Internet or in the financial newspaper.
- III. It should not be possible to consistently profit by trading on private information, such as that obtained from a thorough analysis of the company and its industry.
- IV. It should not be possible to consistently profit by trading on inside information.

(2)

ii) Which behavioural biases are exhibited by following statements?

- a) The investment team of an insurance company decides to use a forward rate agreement with one broker to hedge their portfolio because it is widely used by investment team of other companies. The investment team makes this decision despite academic research suggesting that the new approach of hedging is substantially better.
- b) The car loan interest rates have remained at historically high levels for several years now and there is no compelling reasons for them to change. However, customer A continues to expect them to decrease at some time in the near future.

(2)

(2)

iii) If the intrinsic value of an option is ₹ 10 then

- a. Option is in-the-money
- b. Option is out-of-money
- c. Is a Put option
- d. Is an American call option

Which of the statement is correct?

- I. Only a is correct
- II. Only b is correct
- III. Both b & c are correct
- IV. a, c & d are correct

(2)

- iv) For a 3-month European call option of a non-dividend paying share the current share price is ₹ 104 and strike price is ₹ 100. If the risk-free interest rate is 12% per annum and volatility is 30% per annum then
- The price of this call option is 10.11
 - If the strike price and current share price is halved then the price of call option will be half of the price calculated in (a)
 - If the strike price and current share price is halved then the price of call option will be double of the price calculated in (a)
 - Given information not sufficient to calculate the price of call option

Which of the statement is correct?

- Only d is correct
- both a and b is correct
- both a and c is correct
- only a is correct

(3)

[11]

Q. 3) i) Define iso-elastic. (1)

ii) Which utility function does not exhibit iso-elasticity and why?

- $U(w) = \ln(w)$, $w > 0$
- $U(w) = (w^x - 1)/x$, $w > 0$
- $U(w) = a + bw + cw^2$
- $U(w) = -e^{-aw}$, $a > 0$

(2)

iii) One of the risk averse contestants in the game show “Kaun Banega Crorepati” has utility function as $U(w) = w + dw^2$. The contestant states that the upper limit of wealth where she can use this utility function is $w = \text{Rs } 10,00,000$

a) Determine the value of d in the contestant's utility function. (3)

The contestant wins a prize of Rs 2,50,000 in that gameshow. She is then offered the opportunity to exchange this prize for a larger prize of Rs 6,00,000 if she can answer one more question correctly. However, she will receive no prize at all if she gets the question wrong. She estimates her chances of answering the question correctly to be 50%.

b) Determine whether the contestant should take this opportunity to exchange. (3)

[9]

Q. 4) i) Define the following measures of investment risk:

a) Shortfall probability (1)

b) Value at Risk (VaR) (1)

ii) If $Z_t = 0.75a_t + \sqrt{1 - 0.75^2}b_t$, Where a_t and b_t are independent Weiner process then which of the following is correct

- $E[Z_t] = t$
- $E[Z_t] = 0.75$

III. $V[Z_t] = 0.75$

IV. $V[Z_t] = t$ (3)

iii) A portfolio consists of two assets A and B with the correlation co-efficient between their returns is 0.3. An investor has invested an equal amount of ₹1000 in both of these assets. Given that the daily volatilities of both the assets are 1%, find the 4-day 99% VaR for the Portfolio. (4)

[9]

Q. 5) **i)** What is the difference between hedging and arbitrage? (2)

ii) Using the put-call parity relationship for a non-dividend paying share establish a relationship between the following

a) Delta of a European call option is delta of a European put option.

- I.** Same as
- II.** Higher than
- III.** Lower than
- IV.** Can't be compared with (1)

b) Vega of European call option is Vega of European put option.

- I.** Same as
- II.** Higher than
- III.** Lower than
- IV.** Could be higher or lower than (1)

c) Gamma of European call option is Gamma of European put option.

- I.** Lower than
- II.** Opposite to
- III.** Same as
- IV.** Higher than (1)

iii) Identify and explain the arbitrage opportunity if any, in the following situation.

6-months European put option on a company's share (non-dividend paying) is currently ₹ 5. Given the strike price is ₹ 90, risk-free rate of interest (continuously compounding) is 10% per annum and share price is ₹ 80. (4)

[9]

Q. 6) On 1 January 2016 an insurance company writes a Fasal Bima Yojana policy for the farmer. At the end of each year, the farmer's crop is assessed, and if it is less than 100 tonnes, it is deemed to have failed. If the crop fails for two years in a row the insurance policy pays out Rs. 10 Lakhs and then is immediately terminated. Premiums are Rs. 25000 per month, paid in advance, and there are no expenses. This is the only policy the insurance company writes, and it has initial surplus $U > 0$.

Denote by $\Psi(U, t)$ the probability of ruin by time t , measured in years; and $\Psi(U)$ as the ultimate probability of ruin.

i) Explain whether the following statements are TRUE or FALSE:

- a) $\Psi(U,1) < \Psi(U,1.5)$
 b) $\Psi(U,2.5) = \Psi(U,3.33)$
 c) $\Psi(U,3) < \Psi(U,4)$
 d) $\Psi(U,4) = \Psi(U)$ (4)

PV Sindhu is an Indian shuttler who will be competing at the Olympic Games. She has taken out special insurance which pays out if she is injured. If the injury is so serious that her career has ended, the policy will pay 10 Crores and will be terminated. If she is injured but recovers, the insurance payment will be Rs 1 Crore and the policy will continue.

The insurance company's underwriters believe that the probability of an injury in any year is 0.2, and that the probability of more than one injury in a year can be ignored.

If Sindhu is injured, there is a 75% chance that she will recover. Annual premiums are paid in advance, and the insurance company pays claims at the end of the year.

Assume that this is the only policy that the insurance company writes and that it has an initial surplus of Rs. 1 Crore.

- ii) Calculate the annual premium charged assuming the insurance company uses a premium loading of 30%. (2)

- iii) Determine $\psi(1\text{Crore}, 2)$. (4)
[10]

- Q. 7)** A Bank has issued a zero-coupon bond which is repayable in 5 years' time from now. Upon liquidation, the holders of the bond would be paid after all the depositors are fully paid.

The current value of the Assets is INR 15 crores.

The amount payable in 5 years is INR 3 crores. The value of bank deposit today is INR 6.04 crores.

The continuously compounded risk-free rate is 5% per annum and the volatility of the Assets of the Bank is 25% per annum. The value of deposit is valued at face value accumulated with savings bank interest of 3% p.a.

Find the current value of the zero-coupon bond using Merton model. [7]

- Q. 8)** The table below shows cumulative paid claims and premium income on a portfolio of general insurance policies.

Underwriting year	Development year			Premium Income
	0	1	2	
2017	38420	78110	91000	120,400
2018	31500	79000		117,100
2019	44000			135,500

- i) Assuming an ultimate loss ratio of 90% for underwriting years 2017 and 2018, pick the correct development factors using Bornhuetter-Ferguson approach.

- a) 3.2470, 2.2850
 b) 2.6178, 1.1650
 c) 2.2470, 1.1650

- d) 1.1470, 1.2850 (2)
- ii) Estimate the outstanding claims for this triangle using Bornhuetter-Ferguson approach. (5)
- iii) State the assumptions underlying this estimate. (2)
- [9]

Q. 9) Mr. Shah has decided to invest INR 2 lakh in a Government Scheme. The details of the scheme are given below:

- Any citizen can invest in the scheme
- The term is fixed at 10 years
- Interest rate can take the following rates:
5% with probability 20%,
7% with probability 30%
- Once fixed the interest rate remain same for 10 years.
- The mean rate is 7.35%.

However, Mr. Shah does not currently have the money with him. He decides to borrow from a Bank which requires him to repay the amount along with interest at the end of 10 years. The interest charged at the end of each year is unknown but follows Gamma distribution with expected value of 6.5% and co-efficient of skewness of $\frac{2}{3}$.

The interest charged each year is independent of any of the previous years.

Using normal approximation calculate the probability that the expected amount received from the Government scheme is not sufficient to repay the Bank?

[12]

- Q. 10)** i) Explain what is meant by recombining binomial tree. State the main advantage of using it in modelling the share price movements? (3)
- ii) An investor wants to determine the value of 6-months European put option on a non-dividend paying share using a two-step binomial tree. The current price of the share is ₹550 and the payoff is $\text{Max}(500 - S_T, 0)$ where S_T is the price of the share in 6 months. The investor assumes that for the first period (first 3 months) the share price will increase by 10% or decrease by 5% and the continuously compounding risk-free rate of interest is 2% (per 3 months).
For the next 3 months the expected risk-free rate is assumed to be 3% (per 3 months) and the share price to increase by 20% or decrease by 10%.
- a) Calculate the value of the European put option using two-step binomial tree. (6)
- b) Calculate the value of exotic American put option if the payoff is $\text{Max}(500 - t/6 S_t, 0)$ where S_t is the share price at time t (eg. t is 3 at first-step). (5)
- [14]
